



LECTURE 24

# Tree-Based Models & Ensemble Learning

Nonlinear decision rules, overfitting, bagging, and feature importance.

**75 minutes • Trees → pruning → forests**

Senior + first-year graduate ECE

## Whiteboard exercise: classify by rules

### Prompt

How would you distinguish trumpet vs. piano using only engineered audio features? Build a human-readable flowchart.

### Possible first rule

Is spectral flux high? A piano attack may be transient, while a sustained trumpet frame may have lower flux.

### Possible second rule

Is harmonic structure / spectral envelope consistent with a sustained source? Use MFCCs or spectral shape next.

### Teaching point

A decision tree formalizes “if feature  $j < \text{threshold}$ , go left; else go right” and learns the thresholds from data.

## A decision tree is a recursive partition

- At each node, choose one feature and one threshold that makes the child nodes “purer” in class labels.
- Each root-to-leaf path is an AND of threshold rules.
- In 2-D, repeated threshold splits carve the plane into axis-aligned rectangles.

## Gini impurity: probability of class mixing

$$G = 1 - \sum_{k=1}^K p_k^2$$

### Pure node

[30, 0, 0] → G = 0. One class only.

### Balanced 3-class node

[10, 10, 10] → G = 1 - 3(1/3)<sup>2</sup> = 2/3.

- Candidate splits are scored by weighted child impurity.
- Choose the threshold that produces the largest impurity decrease.

## Entropy and information gain

$$H = - \sum_{k=1}^K p_k \log_2 p_k$$

$$IG = H(\text{parent}) - \sum_m \frac{N_m}{N} H(\text{child}_m)$$

### Gini vs entropy

They usually choose similar splits. Gini is slightly cheaper; entropy has a direct information-theoretic interpretation.

### Engineering habit

Focus on the weighted decrease in impurity — not on memorizing which criterion is “better.” Validate performance.

## Worked split: does this threshold help?

Parent: 20 kick, 20 cymbal

$$G_{parent} = 1 - (0.5^2 + 0.5^2) = 0.5$$

Candidate split:

Left: 18 kick, 2 cymbal

Right: 2 kick, 18 cymbal

$$G_{left} = G_{right} = 1 - (0.9^2 + 0.1^2) = 0.18$$

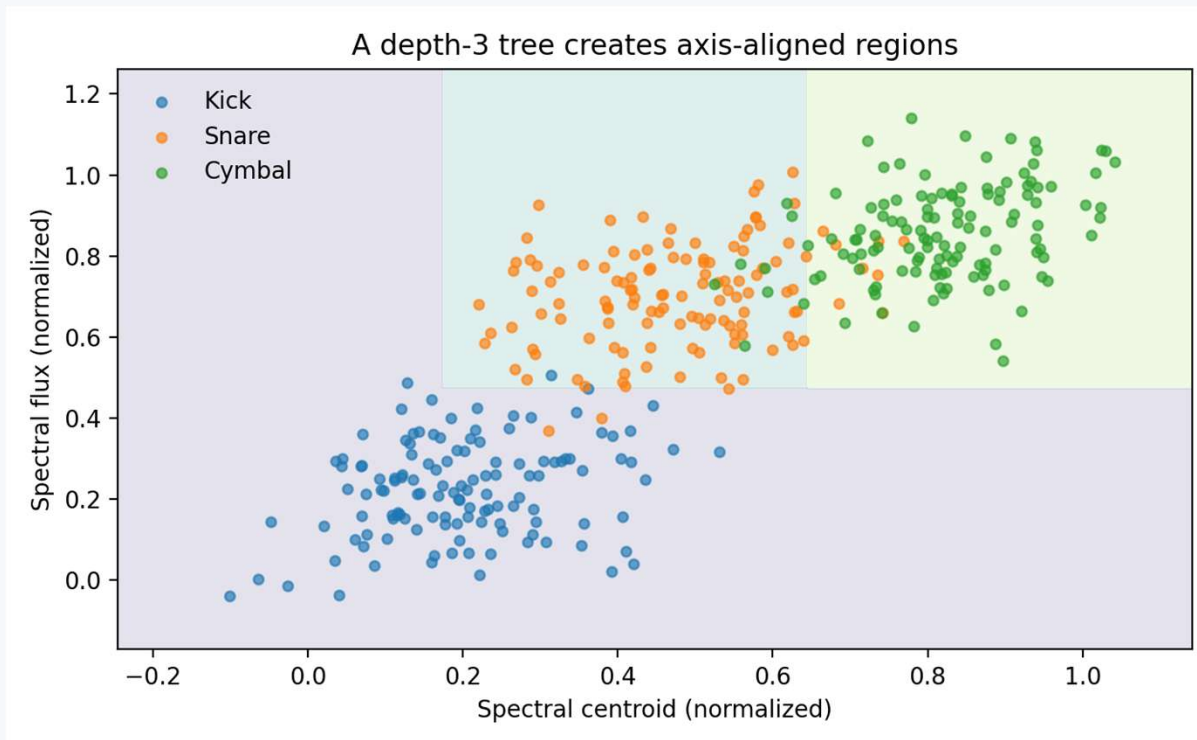
$$G_{weighted} = 0.5(0.18) + 0.5(0.18) = 0.18$$

$$\Delta G = 0.50 - 0.18 = 0.32$$

### Interpretation

Large impurity decrease: the threshold creates two much purer children.

## How a tree partitions audio-feature space



### Axis-aligned boundaries

Each split is perpendicular to one feature axis. Nonlinear overall regions emerge from many rectangular pieces.

### Interpretability

A shallow path can be read as a rule:  $\text{centroid} < \tau_1$  AND  $\text{flux} \geq \tau_2 \rightarrow \text{class}$ .

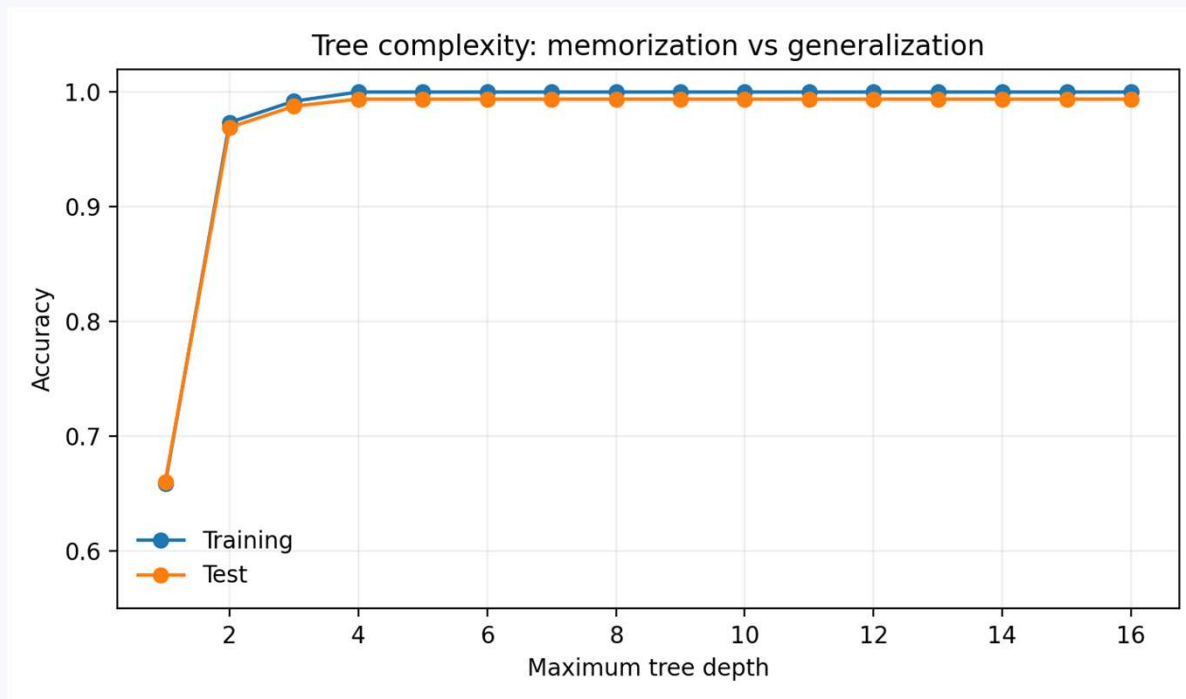
## Trees do not need feature standardization

- A split depends on ordering:  $x_j < \text{threshold}$ . Monotonic rescaling preserves the ordering of samples.
- Therefore standardizing Hz, energy, and MFCC values is not necessary for a standard decision tree.
- But missing values, outliers, categorical encodings, and leakage still require careful preprocessing.



## OVERFITTING

# The danger: a deep tree can memorize the room



### Audio-specific overfit

A deep tree may split on microphone hiss, room noise, gain, performer, or recording-session artifacts rather than instrument identity.

### Symptom

Training accuracy keeps rising while validation/test accuracy stalls or drops.

## Control tree complexity before it controls you

- ``max_depth`` : maximum number of decision levels.
- ``min_samples_leaf`` : require evidence before creating a tiny leaf.
- ``min_samples_split`` : require enough samples to attempt a split.
- ``max_leaf_nodes`` : cap the number of terminal regions.
- ``ccp_alpha`` : post-prune branches using cost-complexity pruning.

## Cost-complexity pruning

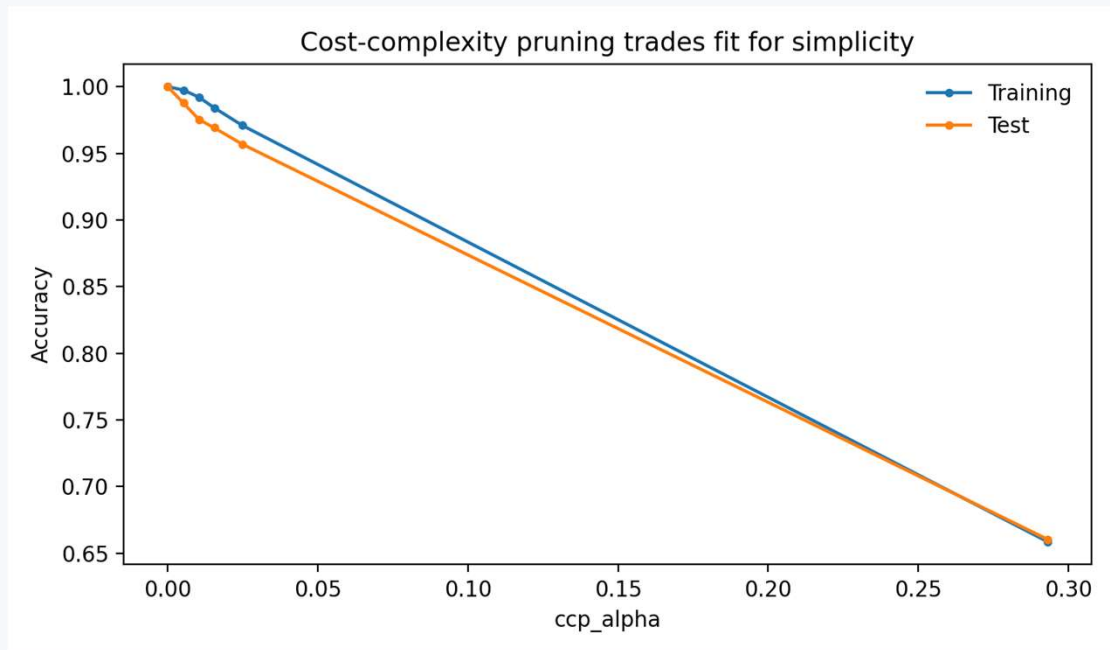
$$R_{\alpha}(T) = R(T) + \alpha|T|$$

**$\alpha = 0$**

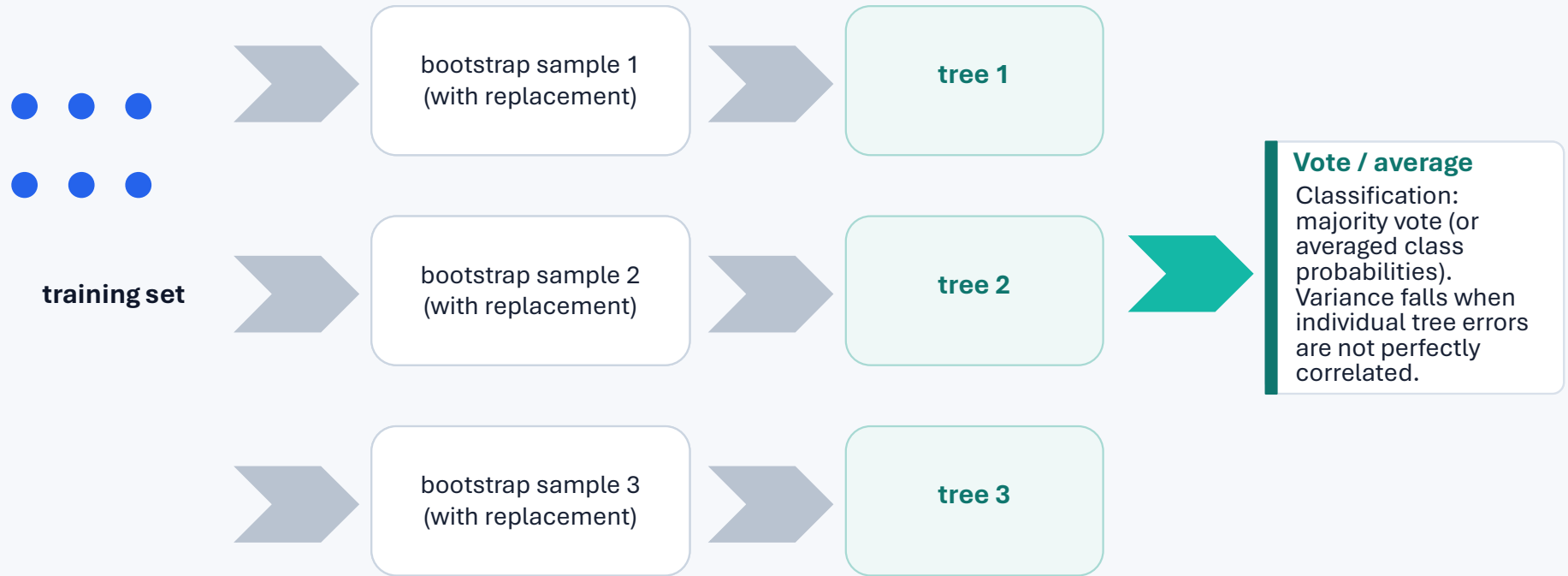
No complexity penalty; retain the fully grown candidate tree.

**Larger  $\alpha$**

Pay more for extra leaves; prune branches that add too little reduction in error / impurity.



## Bagging: average unstable learners



## Random forest = bagging + random feature subsets

- Each tree trains on a bootstrap sample.
- At each split, only a random subset of candidate features is considered.
- This prevents one dominant feature from making every tree look the same.
- Averaging many de-correlated trees typically reduces variance while retaining flexible nonlinear boundaries.

## Why averaging helps

$$\text{Var}(\bar{f}) \approx \rho\sigma^2 + \frac{1 - \rho}{B}\sigma^2$$

**B**

Number of trees. The independent part of variance shrinks roughly with  $1/B$ .

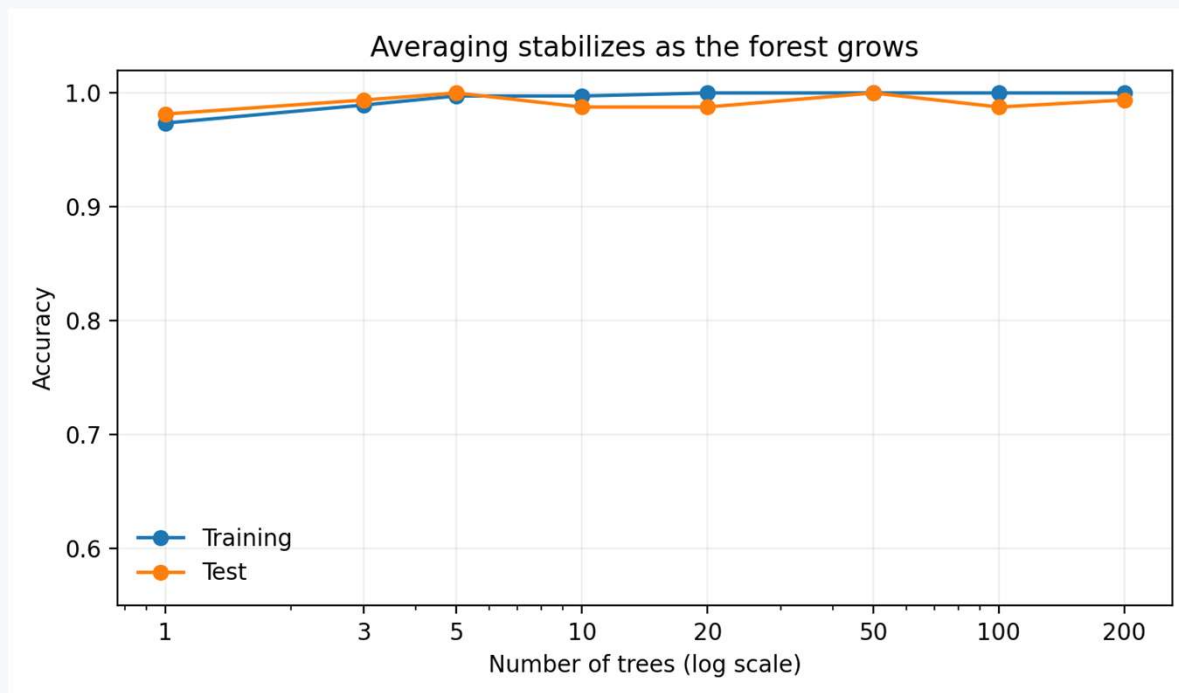
 **$\rho$** 

Correlation between tree errors. If trees make the same mistakes, averaging cannot remove them.

**Design implication**

Bootstrap samples + random feature subsets aim to keep trees strong but less correlated.

## As the forest grows, predictions stabilize



### More trees usually do not overfit like more depth

Increasing  $B$  mainly reduces Monte Carlo variance; training each tree more deeply is a different complexity choice.

### Cost

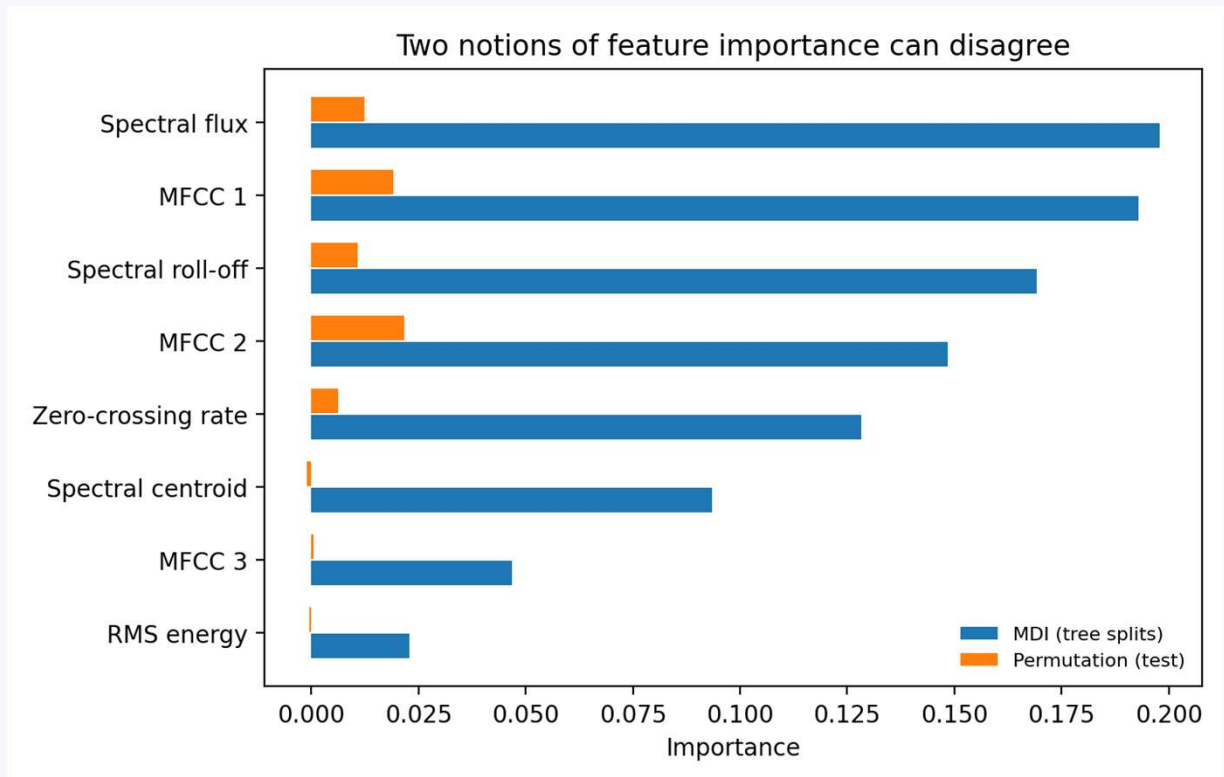
More trees increase memory and inference / training time. Stop when validation performance and uncertainty stabilize.

## Out-of-bag (OOB) evaluation

- A bootstrap sample leaves out roughly 36.8% of the training examples for a given tree.
- For each training example, predict using only trees that did not train on that example.
- Aggregate those predictions to obtain an internal generalization estimate without a separate validation split.



## Feature importance: useful, but ask “importance for what?”



### MDI

Mean decrease in impurity: how much tree splits using a feature reduce impurity across the forest.

### Permutation

Shuffle one feature on held-out data; measure how much prediction degrades.

### Caution

Correlated features can share or mask importance. High importance is not causality.

## Tree vs. forest vs. SVM: choose by structure, not fashion

### Single tree

- Readable rules
- Fast inference
- High variance unless regularized
- Axis-aligned regions

### Random forest

- Flexible nonlinear model
- Robust average
- Good default on tabular features
- Less transparent than one tree

### Scaled SVM

- Margin-based geometry
- Excellent in moderate/high dimension
- Scaling essential
- RBF needs  $C/\gamma$  tuning

**Best engineering answer: compare candidates with the same leakage-safe cross-validation protocol and inspect failure modes.**

## scikit-learn: tree and forest

### Models

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier

tree = DecisionTreeClassifier(
    max_depth=4, min_samples_leaf=8, random_state=1)

forest = RandomForestClassifier(
    n_estimators=300, max_features="sqrt",
    min_samples_leaf=2, oob_score=True,
    random_state=1, n_jobs=-1)
```

### No StandardScaler required

Threshold-based splits are invariant to monotonic rescaling of each feature.

### Notebook

Lecture24\_Trees\_RandomForests.ipynb includes impurity calculations, tree plots, pruning, OOB score, cross-validation, and permutation importance.

## Lecture 24 synthesis

### Tree

Greedy hierarchical threshold rules; intuitive but unstable.

### Regularization

Depth, leaf size, and pruning prevent memorizing recording artifacts.

### Forest

Average many de-correlated trees to reduce variance.

### Interpretation

Importance answers a predictive question, not a causal one; correlated audio features complicate attribution.